

Appendix 9 - Feasibility Risk Register

			Client		Project Name		Scitech Project No		Document Ref		Asymchem		PDF Extension Feasibility		300291		300291-PM-RA-0001		Revision No. A1		Revision Date 21-May-26		By GS		Reason Feasibility Design	
			RISK ENTRY				PRE-ACTION ASSESSMENT (expected outcome without mitigation)																			
Status	Date	REF NO.	RISK	RISK IMPACT	RAISED BY	CATEGORY (from list)	LIKELIHOOD A (SCORE 1-10)	INDEX A /10	PROGRAMME IMPACT (Weeks)	FACTORED PROGRAMME (Weeks)	CRITICALITY (from list)	COST IMPACT	FACTORED COST	RESPONSE (from list)	ACTION OWNER	PROPOSED MITIGATION ACTION										
									182	131.8		£2,636,000	£ 1,757,800													
Open	14/04/2026	1	BREEAM requirements and design-stage implementation	> Missed early-stage opportunities > Project cost increases > Programme delay	MC	Statutory	8	0.8	8	6.4	High	£300,000	£ 240,000	Avoid	GS/CC	> Include allowance in OOM > Include outcome of BREEAM pre-assessment report 12/05/26 in next design stages										
Open	14/04/2026	2	Existing electrical infrastructure may not support extension (LV panel is from 1998), requiring major upgrades.	> Construction cost increase > Feasibility viability impacted > Cost of operational downtime	CG	Extg Site	7	0.7	2	1.4	High	£150,000	£ 105,000	Accept	CG/CC	> Vendor unable to confirm feasibility of modifying existing panel without intrusive survey. Intrusive survey quotation received, requiring full site shutdown. > Modification of existing panel not preferred. Current feasibility-stage options include: 1) Install new panel between existing panel and transformer 2) Split electrical supply arrangement 3) Replace existing main panel with new panel > Feasibility study to proceed on basis of provisional allowance, with preferred solution to be confirmed during next design stage. >Order-of-magnitude (OOM) allowance / contingency to be included within cost plan.										
Open	14/04/2026	3	Limited available space for utilities.	> Increased design & construction scope and cost > Layout iterations and potential re-work > Sub-optimal layout if existing space used	GJ	Extg Site	8	0.8	12	9.6	High	£125,000	£ 100,000	Accept	ST	> Constraints to be clearly highlighted in feasibility report > Develop detailed construction plan > Follow RIBA design stages for increasing definition and accuracy of layout > Potential for process scheduling to reduce extra utility usage in next stages										
Open	14/04/2026	4	Landlord / site-wide constraints limit design and construction options.	> Delays for approvals > Design limitations > Subsequent additional costs	GS	Extg Site	7	0.7	4	2.8	Medium	£40,000	£ 28,000	Mitigate	ST/CC	> Ongoing engagement with landlord in next stages										
Open	14/04/2026	5	Communication barriers between UK and China teams.	> Incorrect assumptions & information > Slower information provision & decisions	GS	Project Org	8	0.8	1	0.8	Low	£10,000	£ 8,000	Mitigate	CC	> Centralise communications, RFI, decision making, meetings etc. via CC > Asymchem to devise strategy to reduce impact of communication barriers										
Open	14/04/2026	6	Late RFI responses.	> Delivery programme slip > Assumptions made to progress design resulting in potential rework	GS	Project Org	7	0.7	2	1.4	Low	£10,000	£ 7,000	Mitigate	GS/CC	> Weekly RFI tracking + escalation + highlight of critical RFIs > Maintain design momentum using assumptions when required > Stage gate approach to design										
Open	20/04/2026	7	Delay in supply chain	> Feasibility programme completion delay	MC	Supply Chain	8	0.8	3	2.4	Medium	£1,000	£ 800	Accept	AW	> Pull estimates from previous projects to produce cost plan > Engage suppliers as early as possible										
Open	20/04/2026	8	Building height increase	> Existing stacks need to be taller > Planning permission delay or denied	JB	Technical	10	1.0	16	16	High	£100,000	£ 100,000	Accept	CC	> Pre-planning to provide feedback on building height > Engage with stack consultant in next stage > Engage with planning consultant in next stage > Stack height & associated surveys increase to be incorporated into cost plan if required										
Open	20/04/2026	9	Increase of equipment lead times	> Delay to construction	MC	Supply Chain	4	0.4	10	4	Medium	£20,000	£ 8,000	Mitigate	AW	> Early order placement > Identification of long lead items and incorporate into procurement programme > Supplier vetting.										
Open	20/04/2026	10	Hydrogenation building cannot be removed	> Redesign of the extension > Increase construction cost as rear wall will need to be blast proof re-inforced concrete	GJ	Technical	5	0.5	2	1	Medium	£200,000	£ 100,000	Accept	CC	> Feasibility design to be based around the removal of hydrogenation building > Two options included for keeping hydrogenation building. > Incorporate selected option into next stage cost plan										
Open	20/04/2026	11	Unforeseen planning permission constraints	> Prolonged planning discussions > Constraints to building design resulting in redesign > Additional cost of planning conditions	GJ	Statutory	5	0.5	16	8	High	£200,000	£ 100,000	Accept	CC	> Pre-planning to provide early feedback on what to provide for planning application										
Open	20/04/2026	12	Inadequate welfare and office facilities	> Modifications required to existing buildings > Increase to construction costs	GJ	Technical	5	0.5	4	2	Medium	£150,000	£ 75,000	Accept	CC	> Statutory requirements reviewed and incorporated into the Feasibility design > Identification of welfare and office requirements at next stage										

POST-ACTION ASSESSMENT (expected outcome with listed mitigation action completed)							
LIKELIHOOD A (SCORE 1-10)	INDEX A /10	PROGRAMME IMPACT (Weeks)	FACTORED PROGRAMME (Weeks)	CRITICALITY (from list)	COST IMPACT	FACTORED COST	NOTES OR FURTHER ACTION
		107	60.6		£1,126,000	£ 626,600	
2	0.2	8	1.6	Low	£20,000	£ 4,000	> Discovery Park (Paul Bax) clarified 'Very-Good' Required for site, however limitations may apply as safety takes precedence. > BREEAM pre-assessment workshop held 11/05/26 and report issued 12/05/26. > Risk reduced but remains live through next design stages pending implementation of identified BREEAM requirements and confirmation of achievable rating.
7	0.7	2	1.4	Low	£0	£ -	> Panel vendor (Andy Thompson, AVK) contacted in Feasibility > Cost impact reduces due to incorporation in OOM
4	0.4	0	0	Medium	£125,000	£ 50,000	> Utilities requirements confirmed with client (current + future demand) via Feasibility RFIs > Available space validated during site review & on drawings (incl. photos + dimensions) and informed Feasibility layout
3	0.3	4	1.2	Low	£40,000	£ 12,000	> Discovery Park (Paul Bax) engaged throughout Feasibility study and site constraints discussed/identified
2	0.2	1	0.2	Low	£10,000	£ 2,000	
10	1.0	1	1	Low	£10,000	£ 10,000	> Design assumptions made at Feasibility study and recorded in report > To be addressed in next design phases.
6	0.6	3	1.8	Low	£1,000	£ 600	> Partially realised by delayed cost receipt for Hastelloy Vessels. To be proceed with informed estimation for Feasibility. > Alternative supplier/ engagement strategy with De Dietrich to be explored in next stage.
10	1.0	2	2	Medium	£40,000	£ 40,000	> Building height increase incorporated into final layouts and Feasibility report.
2	0.2	10	2	Medium	£20,000	£ 4,000	
2	0.2	2	0.4	Low	£0	£ -	> Cost impact reduces due to hydrogenation building removal incorporation in OOM > Feasibility constructability review 14/05/26 validates approach
5	0.5	2	1	Medium	£200,000	£ 100,000	> Risk to be updated after pre-app feedback. > Programme mitigation not cost.
5	0.5	4	2	Medium	£150,000	£ 75,000	> Risk to be updated after decision made at next stage.

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Open	20/04/2026	13	Desired construction completion date of Q3 2028 is tight timeline	> Overall programme increased	GJ	Scope	9	0.9	26	23.4	High	£0	£ -	Mitigate	CC	> Compress overall programme by DNB contract employed at Stage 4 > Programme reduction strategies during construction e.g. 24/7 working	9	0.9	16	14.4	High	£0	£ -	> To be re-assessed in later stages		
Open	20/04/2026	14	Asbestos	> Delay to construction > Cost of removal	GS	Extg Site	8	0.8	6	4.8	Medium	£100,000	£ 80,000	Avoid	ST	> R&D Survey before construction starts (Stage 3)	5	0.5	0	0	Medium	£50,000	£ 25,000			
Closed	21/04/2026	15	Future obsolescence of main switchboards	> Replacement of board > Full shutdown of site - just asymchem building or entirety of discovery park	CG	Extg Site										> Covered by Risk Register Item #2										
Open	20/04/2026	16	Cost increase of equipment during design phase	> Increase capital cost > Project viability	MC	Supply Chain	7	0.7	8	5.6	High	£80,000	£ 56,000	Mitigate	AW	> Cost plans developed in-line with RIBA stage progression with contingencies and assumptions appropriate to design maturity and improving with accuracy as project progresses > Identify long lead/high-risk equipment early	7	0.7	2	1.4	Low	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	20/04/2026	17	Force Majeure	> Delay to project progression at any stage > Associated increase in project costs	GS	External	7	0.7	12	8.4	High	£300,000	£ 210,000	Mitigate	GS/CC	> Maintain and regular review of appropriate design/procurement/construction programme > Project lifecycle split into stage gates (i.e. RIBA stages) controlled project development with clear go/no-go > Monitor supply chaing/procurment risks throughout project lifecycle	7	0.7	12	8.4	High	£300,000	£ 210,000			
Open	24/04/2026	18	Unclear purchasing strategy (responsibly party, timing etc.)	> Delay to procurement > Missed orders > Delay to construction programme	CC	Project Org	8	0.8	12	9.6	High	£80,000	£ 64,000	Avoid	ST/CC	> Early identification of long lead items > Procurement strategy to be detailed in design and construction programme > Deliverable at concept stage that breakdown who owns what > Asymchem to have internal discussion/meeting to devise procurement strategy	4	0.4	4	1.6	Low	£20,000	£ 8,000			
Open	24/04/2026	19	Equipment scope change post-concept	> Redesign/rework cost > Delay to overall programme	CC	Scope	7	0.7	4	2.8	Medium	£40,000	£ 28,000	Mitigate	ST/CC	> Stage gate approach > Asymchem to ensure agreement and alignment with CN stakeholders > ST to follow formal change control process (EWN/CO) > Client progress meetings and workshops to inform design (20%, 50% and 90% reviews in concept/scheme/detailed stages)	4	0.4	4	1.6	Low	£40,000	£ 16,000			
Open	29/04/2026	20	Surface water drainage likely clashes with building	> Reroute required impacting layout > Increased structural and civil scope > Increased programme timeline and project cost	GJ	Extg Site	7	0.7	4	2.8	Medium	£80,000	£ 56,000	Accept	GJ	> Review drainage strategy and co-ordinate with structural/civil consultant in concept stage. > Incorporate into cost plan	7	0.7	4	2.8	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	29/04/2026	21	HV cable location uncertain	> Reroute or building over required > Increased structural and civil scope > Increased programme timeline and project cost	GJ	Extg Site	7	0.7	4	2.8	Medium	£80,000	£ 56,000	Accept	CG	> Confirm HV cable location early and assess diversion/build-over requirements during concept development.	7	0.7	4	2.8	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	06/05/2026	22	Potential requirement to cool third floor due to increased reactor sizes and heat gains.	> Increased utility demand, HVAC sizing and infrastructure requirements > Increased project and construction costs	MC	Scope	6	0.6	4	2.4	High	£150,000	£ 90,000	Accept	JB/AW	> Include cooling for entire building in cost plan and value engineer at later stage > Client to confirm cooling requirements	6	0.6	4	2.4	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	21/04/2026	23	Existing reactor cooling philosophy may not be suitable for proposed larger-scale reactors.	> Later design stages may require redesign of reactor cooling systems, increased utility capacity, revised controls strategy > Additional capital cost and programme delay.	GS	Extg Site	5	0.5	4	2	Medium	£80,000	£ 40,000	Accept	JB	> Feasibility study proceeding on the assumption that the existing crash cooling philosophy is suitable for the proposed larger scale reactors. > Reactor cooling philosophy to be assessed and validated next stage.	5	0.5	4	2	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	21/04/2026	24	Existing emergency power philosophy and generator capacity may not be sufficient for proposed process expansion.	> Additional generator capacity, new essential boards or revised backup philosophy may be required. > Increase in capex and programme.	GS	Extg Site	5	0.5	2	1	Medium	£80,000	£ 40,000	Mitigate	CG	> Feasibility based on installation of a new dedicated generator for the extension. > Cost to be included in OOM	5	0.5	2	1	Low	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	21/04/2026	25	Existing waste handling arrangements and transfer line capacity may not support proposed process expansion.	> If unavailable, increased tanker movements, additional storage tanks, operational constraints. > Higher opex/capex may result.	GS	Extg Site	5	0.5	4	2	Medium	£80,000	£ 40,000	Accept	MC/GJ	> Feasibility assumes existing waste handling arrangements can be maintained or expanded, including potential reinstatement of transfer line to Waste Care. > Waste handling arrangements to be assessed and validated next stage	5	0.5	4	2	Medium	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	21/04/2026	26	Existing nitrogen, compressed air, thermal fluid and cooling systems may not have sufficient spare capacity for proposed extension.	> Discovery of insufficient capacity may require new plant, enlarged distribution systems and redesign.	GS	Extg Site	7	0.7	4	2.8	Medium	£80,000	£ 56,000	Mitigate	MC	> Preliminary calcs have estimated new nitrogen, hot thermal fluid heating and distribution new cooling utilities are required. > To be incorporated into next stage design.	7	0.7	0	0	Low	£0	£ -	> Cost impact reduces due to incorporation in OOM		
Open	21/04/2026	27	Unknown foundation and structural design.	> Later intrusive surveys may identify abnormal ground conditions, foundation complexity or structural constraints increasing cost.	GS	Extg Site	7	0.7	8	5.6	High	£100,000	£ 70,000	Accept	GJ	> Feasibility assumes that similar design to existing site can be used. > Structural/Civil Engineer and Ground Investigation Consultants to be instructed in next stage > To be allowed for in OOM	7	0.7	8	5.6	High	£100,000	£ 70,000			